

Speech Recognition — Metrics Summary

Benchmark of local Persian STT models (FLEURS fa_ir), [plus the clinical metrics used to evaluate dictated radiology reports](#). Why each metric is tracked.

■ Blue = added for clinical (radiology) evaluation. Black = the original STT benchmark set.

Accuracy

Metric	Why we use it
WER	% of words wrong — the standard ASR headline metric.
CER	% of characters wrong — fairer for Persian's attached suffixes/clitics.
chrF	Character n-gram overlap — gives partial credit for morphological variants WER counts as fully wrong.
BERTScore (ParsBERT)	Token-level <i>meaning</i> match — credits valid synonyms that WER penalizes.
Semantic similarity	Whole-sentence meaning match via embeddings — did it preserve the message.

Reliability (critical for medical use)

Metric	Why we use it
SER	% of clips with <i>any</i> error — how often a transcript is perfect.
WER P50 / P90 / P95	Typical vs worst-case clips; averages hide bad outliers.
% perfect / % catastrophic	Share of flawless vs badly-wrong clips.
Sub / Ins / Del rates	Error <i>type</i> ; insertions = hallucinated words (highest medical risk).
Empty-output rate	How often the model returns nothing.

Persian-specific failure modes

Metric	Why we use it
Hallucination ratio	Output vs reference length; >1 means inventing words.
Script contamination	% non-Persian characters — flags Latin/English leakage.
Repetition score	Repeated n-grams — catches Whisper's loop-hallucination on silence.
Punctuation F1	Fairly compares punctuating models (Whisper) vs raw-text ones (Wav2Vec2/MMS).

Does not transfer to dictated radiology. Script contamination assumes Latin characters are leakage. In radiology dictation the speaker mixes Persian with English terms on purpose (*hydronephrosis*, *UVJ*, *mm*), so a correct transcript scores as heavily contaminated. Treat it as a FLEURS-only metric, or replace it with a code-switching check that scores whether English terms were rendered correctly rather than whether they appear at all.

Clinical accuracy (dictated radiology reports)

Scored on the report text, against a radiologist-verified reference. These are the metrics that carry clinical risk — a report can have a good WER and still be unsafe.

Metric	Why we use it
Negation error rate	Presence/absence flips — "no stone" transcribed as "stone". Changes the diagnosis outright.
Laterality error rate	Right/left flips (راست/چپ, Right/Left) — sends a finding to the wrong side of the body.
Number error rate	Wrong measurement values — the single highest-risk error class; a 6 mm stone read as 1 cm changes management.
Unit error rate	mm vs cm vs میلی‌متر vs سانتی‌متر. Compare physical quantities, not unit strings, so 10 mm and 1 cm are equal.
Medical term precision	Of the clinical terms produced, how many are actually supported by the reference — catches invented terminology.
Medical term recall	Of the clinical terms that should be present, how many survived — catches dropped findings.
Medical term F1	Single combined figure for terminology handling.
Critical omission rate	Important reference content missing from the output. "Critical" must be defined explicitly (measurement, negated finding, laterality marker, or flagged term) or it collapses into "anything missing".
Unsupported addition rate	Output content with no support in the reference — the fabrication case (e.g. a model inventing a patient history that was never dictated).
requires_medical_review	Per-report flag raised by any critical error. In a post-edit workflow this fires <i>after</i> sign-off, so it is an audit trigger for a second look, not a safety gate.

Post-edit loop (live monitoring)

When the reference is the radiologist's correction of our own output, the numbers mean something different: they measure editing effort, not correctness.

Metric	Why we use it
Edit burden (post-edit WER)	How much the radiologist had to change. A production workload signal — do not read it as accuracy, since the reference is derived from the model's own text.
% accepted unchanged	Share of reports signed with no edit. Track alongside engagement signals: a rubber-stamped report is indistinguishable from a correct one.
Correction return rate	Share of outputs that ever come back corrected. Low or skewed return means the evaluated sample is biased toward whichever reports are easy to edit.